

CHAPTER 8

8.1) HEAT PUMP

(a) If operating reversibly, this is just a Carnot engine with all flows reversed, hence (7) remains valid. As in a refrigerator, any entropy generated in a heat pump must leave at the higher temperature τ_h . Hence, as in (14),

$$Q_h/\tau_h = \sigma_h \geq \sigma_\ell = Q_\ell/\tau_\ell \quad ,$$

where the equals sign holds in the limit of reversible operation. Solved for Q_ℓ :

$$Q_\ell = \tau_\ell \sigma_\ell \leq \tau_\ell \sigma_h = (\tau_\ell/\tau_h) Q_h \quad .$$

Hence the energy required to drive the heat pump is

$$W = Q_h - Q_\ell \geq (1 - \tau_\ell/\tau_h) Q_h = \eta_C Q_h \quad . \quad (S1)$$

The reversible limit of this is equivalent to the claimed result; for irreversible operation more work is required.

(b) From (6), with τ_h and Q_h replaced by τ_{hh} and Q_{hh} :

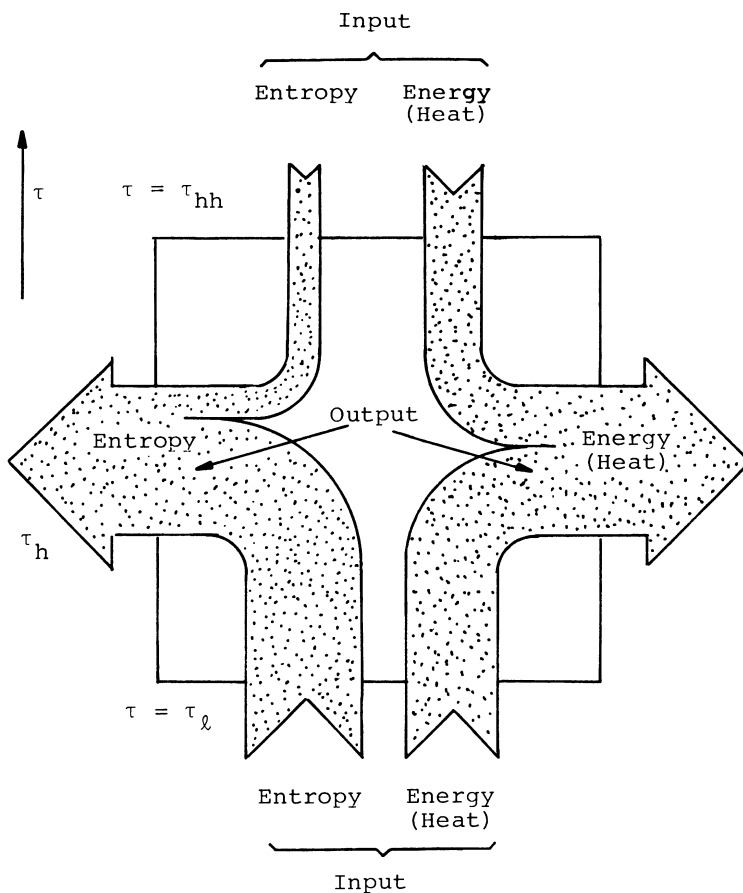
$$W = (1 - \tau_\ell/\tau_{hh}) Q_{hh} \quad .$$

Inserted into the reversible limit of (S1):

$$\begin{aligned} (1 - \tau_\ell/\tau_{hh}) Q_{hh} &= (1 - \tau_\ell/\tau_h) Q_h \quad , \\ Q_{hh}/Q_h &= (1 - \tau_\ell/\tau_h) / (1 - \tau_\ell/\tau_{hh}) \\ &= (\tau_{hh}/\tau_h) (\tau_h - \tau_\ell) / (\tau_{hh} - \tau_\ell) = \eta_C(\tau_h, \tau_\ell) / \eta_C(\tau_{hh}, \tau_\ell) \quad . \end{aligned}$$

For our numerical example, $Q_{hh}/Q_h \cong 0.18$.

(c) See next page.



8.2) ABSORPTION REFRIGERATOR

(a) See figure above, which applies to this problem as well.

(b) From energy and entropy conservation:

$$Q_h = Q_{hh} + Q_{\ell} \quad , \quad \sigma_h = Q_h/\tau_h = \sigma_{hh} + \sigma_{\ell} = Q_{hh}/\tau_{hh} + Q_{\ell}/\tau_{\ell}.$$

Multiply the entropy relation by τ_h and subtract from the energy relation:

$$\begin{aligned}
0 &= (1-\tau_h/\tau_{hh})Q_{hh} + (1-\tau_h/\tau_\ell)Q_\ell \quad ; \\
Q_\ell/Q_{hh} &= (1-\tau_h/\tau_{hh})/(\tau_h/\tau_\ell - 1) \\
&= (\tau_\ell/\tau_{hh})(\tau_{hh}-\tau_h)/(\tau_h-\tau_\ell) = \eta_C(\tau_{hh}, \tau_h) \gamma_C(\tau_h, \tau_\ell) \quad .
\end{aligned}$$

A detailed description of the actual operating cycle of gas-heated absorption refrigerators is given in the Supplementary Material to this Chapter.

8.3) PHOTON CARNOT ENGINE

This problem draws upon (4.2) and (4.23), which we write in the abbreviated form

$$U = sV\tau^4 \quad , \quad \sigma = 4sV\tau^3/3 \quad , \quad (4.20, 23)$$

where $s = \pi^2/15\hbar^3 c^3$. We designate by W_{ij} and Q_{ij} the work taken up by the system in going from "i" to "j".

(a) According to (4.23), $V\tau^3 = \text{constant}$ in an isentropic process. Point 3 on the Carnot loop is isentropic with point 2 (see Figure 4.5), and point 4 is isentropic with point 1; hence

$$V_3\tau_\ell^3 = V_2\tau_h^3 \quad \text{or} \quad V_3 = V_2 \times (\tau_h/\tau_\ell)^3 \quad , \quad (S1)$$

$$V_4\tau_\ell^3 = V_1\tau_h^3 \quad \text{or} \quad V_4 = V_1 \times (\tau_h/\tau_\ell)^3 \quad . \quad (S2)$$

(b) From $\delta Q = \tau d\sigma$, with (4.23):

$$Q_h = Q_{12} = \tau_h \int_1^2 d\sigma = \tau_h \times (\sigma_2 - \sigma_1) = (4s\tau_h^4/3)(V_2 - V_1) \quad . \quad (S3)$$

Also, from $\delta W = dU - \delta Q$, with (4.20):

$$\begin{aligned}
W_{12} &= U_2 - U_1 - Q_{12} = s\tau_h^4 \times (V_2 - V_1) - Q_{12} \\
&= - (s\tau_h^4/3)(V_2 - V_1) = - Q_h/4 \quad .
\end{aligned}$$

Hence, heat taken up and work done are not equal.

For the isothermal step $3 \rightarrow 4$ at $\tau = \tau_\ell$ we find similarly

$$W_{34} = -(s\tau_\ell^4/3)(V_4 - V_3) = (s\tau_\ell\tau_h^3/3)(V_2 - V_1) \quad ,$$

where we have utilized (S1,2) to replace V_3 and V_4 by V_2 and V_1 . The combined work done on the system during the two isothermal stages is

$$W_{12} + W_{34} = -(s\tau_h^3/3)(\tau_h - \tau_\ell)(V_2 - V_1) \quad . \quad (S4)$$

(c) No! We calculate the isentropic work as an energy difference, using (4.20), (S1) and (S2):

$$W_{23} = U_3 - U_2 = s \times (V_3\tau_\ell^4 - V_2\tau_h^4) = -sV_2\tau_h^3(\tau_h - \tau_\ell) \quad ,$$

$$W_{41} = U_1 - U_4 = s \times (V_1\tau_h^4 - V_4\tau_\ell^4) = sV_1\tau_h^3(\tau_h - \tau_\ell)$$

$$W_{23} + W_{41} = -s\tau_h^3(\tau_h - \tau_\ell)(V_2 - V_1) \neq 0 \quad . \quad (S5)$$

(d) Total work done by the system, from (S4) and (S5):

$$W = -(W_{12} + W_{23} + W_{34} + W_{41}) = (4s\tau_h^3/3)(\tau_h - \tau_\ell)(V_2 - V_1) \quad .$$

Divided by Q_h from (S3):

$$\eta = W/Q_h = (\tau_h - \tau_\ell)/\tau_h = \eta_C \quad .$$

8.4) HEAT ENGINE-REFRIGERATOR CASCADE

Taking the last question first, the answer to it is clearly: No. The cascade is simply another more complicated heat engine in its own right, which is subject to the Carnot limit just as the individual heat engine and refrigerator are.

The heat engine operates between the temperatures τ_h and τ_r . It is instructive to go beyond the problem statement and to include irreversibility. In this case, from (9) and (10), with the substitution $\ell \rightarrow r$,

$$Q_r \geq (\tau_r/\tau_h)Q_h \quad , \quad (S1)$$

$$W_e = Q_h - Q_r \leq (1-\tau_r/\tau_h)Q_h \quad , \quad (S2)$$

where the equals sign holds in the reversible limit.

For the refrigerator, τ_ℓ is actually the upper temperature and τ_r the lower temperature. If we substitute first $\ell \rightarrow r$, then $h \rightarrow \ell$ in (15) and (16) we obtain

$$Q_\ell \geq (\tau_\ell/\tau_r)Q_r, \\ W_r = Q_\ell - Q_r \geq (\tau_\ell/\tau_r - 1)Q_r \geq (\tau_\ell/\tau_h - \tau_r/\tau_h)Q_h, \quad (S3)$$

where in the last inequality we utilized (S1). The work generated by the cascade is, with (S2) and (S3),

$$W = W_e - W_r \leq (1-\tau_r/\tau_h)Q_h - (\tau_\ell/\tau_h - \tau_r/\tau_h)Q_h = (1-\tau_\ell/\tau_h)Q_h \quad ,$$

which is the expected overall Carnot relation equivalent to (11) for the cascade.

8.5) THERMAL POLLUTION

Let η be the energy conversion efficiency defined by (11), not necessarily equal to the Carnot efficiency. From $W = \eta Q_h = \eta \times (W + Q_\ell)$ we obtain

$$W = [\eta/(1-\eta)]Q_\ell \quad . \quad (S1)$$

For reversible operation, $\eta = \eta_C = (\tau_h - \tau_\ell)/\tau_h$; then

$$W = [(\tau_h - \tau_\ell)/\tau_\ell]Q_\ell \quad . \quad (S2)$$

Numerically, for our example:

$$T_h = 500^\circ\text{C}: W = (480/293) \times 1500 \text{ MW} \cong 2460 \text{ MW} \quad ,$$

$$T_h = 600^\circ\text{C}: W = (580/298) \times 1500 \text{ MW} \cong 2970 \text{ MW} \quad .$$

Because of unavoidable irreversibilities, the actual capacity will be less, not only because for a given amount of Q_h less power will be generated, but Q_h must actually be reduced to keep Q_ℓ fixed.

Comment. The factors preceding Q_ℓ in (S1) and (S2) are just the inverses of the coefficient of refrigerator performance, in the general case and in the Carnot limit. Thus we see that their utility is not restricted to refrigerators, but extends to other cases in which the relation between W and Q_ℓ is of interest.

8.6) ROOM AIR CONDITIONER

(a) We have $dQ_\ell/dt = A \times (T_h - T_\ell)$. From this and (12):

$$P = dW/dt = [(T_h - T_\ell)/T_\ell] dQ_\ell/dt = A \times (T_h - T_\ell)^2 / T_\ell \quad (S1)$$

This may be re-arranged into a quadratic equation for T_ℓ :

$$T_\ell^2 - (2T_h + P/A)T_\ell + T_h^2 = 0 \quad .$$

Its relevant solution ($T_\ell < T_h$) is

$$T_\ell = (T_h + P/2A) - [(T_h + P/2A)^2 - T_h^2]^{1/2} \quad .$$

(b) From (S1):

$$A = PT_\ell / (T_h - T_\ell)^2 = 2 \text{ kW} \times 290 \text{ K} / (20 \text{ K})^2 = 1.45 \text{ kW K}^{-1} \quad .$$

8.7) LIGHT BULB IN A REFRIGERATOR

Yes: We have $Q_\ell = W$. If this is inserted into (12) one obtains $T_h - T_\ell = T_\ell$ or $T_\ell = T_h/2 \cong 150 \text{ K}$.

8.8) GEO THERMAL ENERGY

The work δW extractable from the heat δQ_h by a Carnot engine is, from (10),

$$\delta W = [(T_h - T_\ell)/T_h] \delta Q_h = -MC(1 - T_\ell/T_h) dT_h \quad ,$$

where T_h is the temperature at which δQ_h is removed from the rock. The total work extractable follows by integration over T_h :

$$W = - MC \int_{T_i}^{T_f} (1 - T_\ell / T_h) dT_h = - MC [T_f - T_i - T_\ell \log(T_f / T_i)] .$$

Numerically, $MC = 10^{17} \text{ J K}^{-1}$. The value of the brackets is -249 K. With these numbers:

$$W = 10^{17} \text{ J K}^{-1} \times 249 \text{ K} / (3.6 \times 10^6 \text{ J/kWh}) = 6.92 \times 10^{12} \text{ kWh} .$$

8.9) COOLING OF NON-METALLIC SOLID TO $T = 0$.

Let $dQ_\ell = -CdT_\ell = -aT_\ell^3 dT_\ell$ be the heat removed from the solid in lowering its temperature from T_ℓ to $T_\ell + dT_\ell$ (we assume $dT_\ell < 0$). The reversible work to do so is, from (12):

$$dW = [(T_h - T_\ell) / T_\ell] dQ_\ell = - a [T_h T_\ell^2 - T_\ell^3] dT_\ell .$$

The total work required to cool the solid from $T_\ell = T_h$ to $T_\ell = 0$ follows by integration

$$\begin{aligned} W &= - a \int_{T_h}^0 [T_h T_\ell^2 - T_\ell^3] dT_\ell \\ &= a \left[\frac{1}{3} T_h T_\ell^3 - \frac{1}{4} T_\ell^4 \right]_0^{T_h} = \frac{1}{12} a T_h^4 . \end{aligned}$$

Comment. The Third Law makes it fundamentally impossible to build any refrigerator that reaches $\tau = 0$ in a finite number of cycles: Because the entropies of all systems go towards a constant value as $\tau \rightarrow 0$, $d\sigma/d\tau \rightarrow 0$, the heat capacity of the working substance of a Carnot refrigerator goes to zero more rapidly than τ itself. But this means that the heat removable per cycle goes to zero more rapidly than τ itself: Zero absolute temperature can be achieved only asymptotically, after an infinite number of cycles. Because of this consequence, the third law is some-

times referred to as the law of the unattainability of absolute zero.

In all real refrigerators the attainable temperature is limited by the entropy generated per cycle due to unavoidable irreversibilities. When that entropy reaches the entropy removed per cycle, further cooling ceases.

8.10) IRREVERSIBLE EXPANSION OF A FERMI GAS

The average energy per particle remains unchanged. Initially, from (7.10) and (7.7):

$$\langle \epsilon \rangle = 3\epsilon_F/5 = (3\hbar^2/10M)(3\pi^2 N/V_i)^{2/3} .$$

After expansion to a concentration $n \ll 2n_Q$ the gas behaves like an ideal gas, with $\langle \epsilon \rangle = 3\tau/2$. Equating the two values:

$$\tau = 2\epsilon_F/5 = (2\hbar^2/10M)(3\pi^2 N/V)^{2/3} . \quad (S1)$$

The condition $n \ll 2n_Q$ can be written, with (3.63):

$$N/V_f \ll 2n_Q = 2(M\tau/2\pi\hbar^2)^{3/2} .$$

Inserting τ from (S1):

$$N/V_f \ll (3/5)(\pi/10)^{1/2}(N/V_i) \text{ or } V_f/V_i \gg (5/3)(10/\pi)^{1/2} \cong 3 .$$

Thus any expansion by a factor large compared to 3 will do.

(a) 7860 K. (b) 9.22×10^5 K.

SUPPLEMENTARY MATERIAL TO CHAPTER 8

ENERGY CONVERSION VERSUS ENERGY STORAGE

The instructor may wish to go beyond the text in stressing that the Carnot inequality applies to the conversion of heat into work in a cyclical device, but not to the temporary storage of work as heat in a one-shot process. One can store work as heat and then retrieve all the heat as work, for example by the isothermal compression and expansion of an ideal gas. Because the energy of an ideal gas at constant τ is independent of the volume, all the work performed on the gas in compression is transferred as heat to the temperature reservoir. On expansion, all the heat is retrieved as work. No mechanical efficiency is lost if the gas is not ideal: the change in internal energy on compression is regained on expansion. A compressed-air tank is an example of such an energy storage device. Strictly speaking, it stores free energy F rather than energy U : Its temperature is kept (more or less) constant by heat exchange with its environment. Hence it stores isothermal work. But we saw in (36) that isothermal work is equal to the change in Helmholtz free energy F of the system.

COOLING WITH A FLAME: ABSORPTION REFRIGERATOR

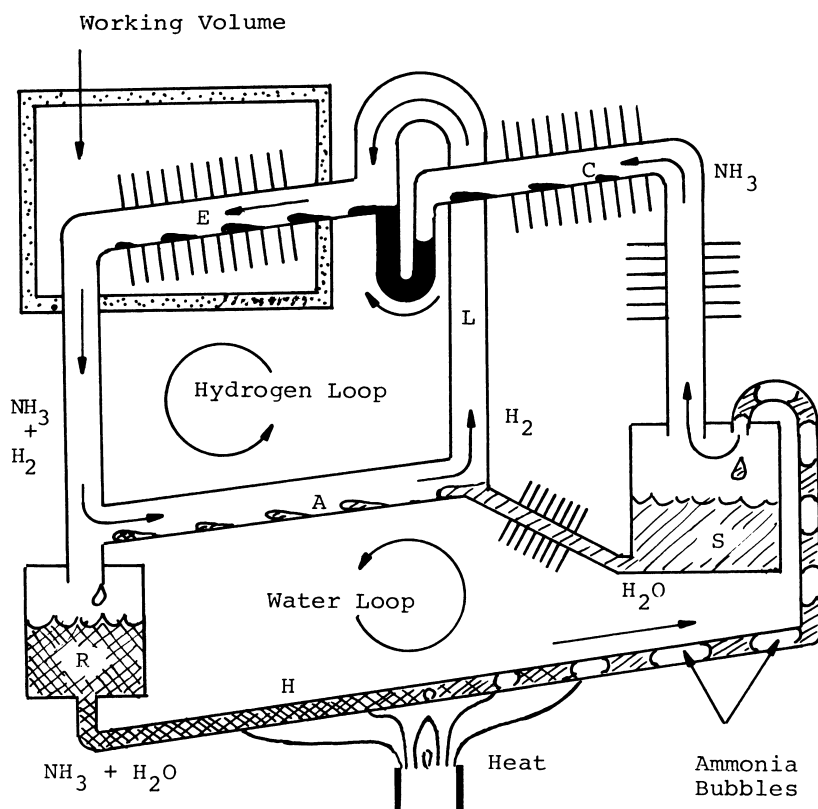
Students often express curiosity about the exact operating cycle of refrigerators that do not possess any moving mechanical parts and in which the driving energy is provided by the heat of a burning flame. The possibility of such refrigerators was analyzed in Problem 8.2. We discuss here the cycle most commonly employed in these refrigerators, in engineering textbooks usually called the Electrolux cycle, after the Swedish company that developed it.

The overall arrangement is shown in the drawing on the next page. A reservoir R contains concentrated ammonia-water solution. Inside a heater pipe H ammonia is boiled out of the water, forming vapor bubbles in a percolator pipe P, just as in a coffee percolator. Because of the reduced average density of the bubble train, this train rises to a height above the liquid level in the reservoir R, spilling into a separator S, where the ammonia vapor and the liquid water separate. The ammonia vapor retains some water vapor, and the water retains some dissolved ammonia, but these complications do not affect the operating principle.

From the separator S the ammonia vapor flows into a condenser C. The condensed ammonia spills over, through a U-tube, into an evaporator E, where the liquid ammonia takes up heat from the working space of the refrigerator, evaporating in the process. The still-cold ammonia vapor then flows into an absorber A.

The water accumulating in the separator also eventually spills over into the absorber, where it meets with the cold ammonia vapor. In the absorber the still-cold ammonia vapor both cools the water and gets re-absorbed by the water, reconstituting the concentrated ammonia-water solution in the reservoir R. Refrigerators in which the cyclic absorption and desorption of a gas are utilized in the cooling cycle are called absorption refrigerators.

There is one problem with the arrangement described so far: Because of the different temperatures in the condenser and evaporator, the equilibrium vapor pressures of ammonia in the two regions differ drastically. The total pressure throughout the system is equilibrated by adding an appropriate amount of an inert gas (hydrogen) on the cold side. During manufacture, the entire system is simply filled with a certain amount of hydrogen. Once the



refrigerator is turned on, most of the hydrogen gets purged automatically from the hot side through the U-tube as a temperature difference develops between the two half-loops. The overall distribution of hydrogen automatically adjusts itself in such a way that the total pressure in both half-loops is very nearly the same.

The operation of the refrigerator is improved by adding a hydrogen return loop L. The cold ammonia-rich hydrogen-ammonia mixture at the lower end of the evaporator is much denser than the warm ammonia-poor hydrogen at the upper end of the absorber. The path L then establishes a hydrogen circulation, which serves two functions:

a) It aids in sweeping the ammonia vapor from the evaporator, thereby speeding up the evaporation rate and hence the heat take-up rate.

b) It cools the return water in the absorber. This reduces the equilibrium ammonia pressure of the ammonia-water solution, thereby facilitating the re-absorption of the ammonia by the water.

The purpose of the cooling fins on the path S-C is to remove as much water vapor as possible from the ammonia vapor before condensing the ammonia itself, at least partially to prevent ice from plugging up the evaporator. The role of the cooling fins on the path S-A should be obvious.

From a Carnot point of view, the Electrolux refrigerator is a low-performance device. To boil the ammonia out of the water solution, water has to be heated along with the ammonia. The heating and subsequent cooling of this water places a double energetic burden on the cycle. Still, it works, and with its utter simplicity and absence of any moving mechanical parts it is very reliable.